

Testing Digital Twin Control Strategies On A Physical Twin: Long-term Evaluation

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Evaluating Digital Twin Control Strategies on a Physical Twin: Test for Reliability

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Abstract: Most water production systems are still operated in a steady, non-adaptive way. However, increasing pressures to remove an ever-increasing set of pollutants (e.g., virus removal, PFOAs), to reduce environmental impact, and to reduce both capital and operational costs make the use of automated or even fully autonomous adaptive operations an attractive proposition (Conejos Fuertes et al. 2020, Ghorbani Bam et al. 2025). In this work, we evaluate a model predictive control (MPC) strategy to optimize the production schedule of the OCWD water treatment system for potable reuse.

Keywords: Cost minimization; Energy efficiency; Desalination; Digital twin; Optimal production scheduling

BACKGROUND AND PROBLEM DEFINITION

Since the advent of water quality sensor technology in the 1980's, water production systems are instrumented with increasing numbers of sensors. Today, water utilities easily produce sensor signals in scale of 1000s. Historically, municipal wastewater treatment plants have invested routinely in control technology to handle diurnal and seasonal dynamics in their load. In contrast, water production systems have been shielded from such dynamics so far by means of buffer tank capacity and fixed rate resource prices (e.g., for electricity). In anticipation of changing needs and opportunities for advanced control, we test a digital twin for adaptive optimization of the flow rate schedule to account for hourly changes in the electricity tariff. To this end, we build a data-driven model in Python and use it to optimize setpoints for the permeate, reject (brine), and recirculation flow rates on an hourly basis using a 7-day forecast. We tested this strategy on a physical twin, a pilot mimicking the full-scale plant, during a period of 2 weeks.

MATERIALS AND METHODS

Physical twin. Figure 1 shows the pilot. It includes three units for (a) ultrafiltration, (b) reverse osmosis, and (c) ultraviolet and advanced oxidation treatment. In this work, we manipulate the flow rates in the RO unit, specifically by changing the permeate, concentrate, and recirculation flow rates. Note that these three flow rates determine all other flow rates (membrane feed, membrane reject, unit make-up). For this work, the sensor signals of interest are (a) the permeate, reject (brine), and recirculation flow rate measurements and (b) the aggregate power demand.

Digital twin. A model of the physical twin was built in Python. To this end, a cubic polynomial regression model was built to predict the power demand given values for the manipulated flow rates. This model has at most 19 parameters and is therefore prone to overfitting. To prevent this, we drive a LASSO penalty on the regression coefficients which drives the regression coefficients to zero (Tibshirani 2011; Fonti and Belitser 2017). To determine the optimal non-zero coefficients, we apply cross-

validation. The resulting, sparse model captures the most important relationships between flow rates, salinity levels, and power demand for the UF, RO, and UV-AOP units of the system. Most importantly, this model enables forecasting of the anticipated power demand for the complete pilot given setpoints for the permeate, reject (brine), and recirculation flow rates. To ensure that the model always delivers reliable predictions, the model is trained (also: calibrated) every hour by the most recent 7-day window (estimation horizon) of sensor data obtained on the pilot, using a least-squares objective function.

This model is then used to optimize the flow rate setpoints. This optimization is focused on minimizing the specific energy cost (SEC) to produce a user-defined volume of permeate during a user-defined optimization horizon, while satisfying equipment and safety constraints. To compute this SEC, we rely on a known tariff schedule. This tariff schedule results in a cost that is proportional to the total power used during a given hour. The proportional cost rate however changes from one hour to the next. The tariff schedule follows a repeatable weekly pattern with changes only twice per year (winter/summer). The optimization problem is implemented to match this tariff schedule by tuning the flow rate setpoints for permeate, reject (brine), and recirculation flow rates for every hour in a 7-day optimization horizon. This leads to a $7 \times 24 \times 3 = 504$ -dimensional optimization problem. This is solved every hour using the most recent model obtained after training. Then, the first setpoints for each flow rate are implemented on the pilot while new sensor signal data are being recorded. Figure 2 illustrates the rolling window approach that delivers hourly updates to the digital twin, including model parameters, as well as the optimal setpoint sequence for the 7-day optimization horizon. The setpoints corresponding to the first hour in the sequence are implemented on the pilot.

RESULTS

Figure 3 illustrates the solution to the optimization at a single time, 1 day after the start of our 2-week test. One can see that the hourly cost with the optimized flow rate setpoints is generally lower (higher) when the electricity tariff is high (low). Note that the optimized flow rates correspond to the same total production volume over the 7-day optimization horizon as would be expected with the default, non-dynamic flow rate setpoints. The first of the optimized setpoints are sent to the programmable logic controller (PLC) of the pilot. However, the specific energy cost is reduced with 38% when applying the optimized flow rate setpoints.

Figure 4 shows the setpoints that are applied by repeating the model calibration and setpoint optimization on an hourly basis. Here, one can see that setpoints are generally lower (higher) when the tariff is high (low). This leads to an effective optimization of the pilot's operation and enables quantification the operational cost savings if a similar approach would be adopted at full scale (in progress). Most importantly, the two-week experiment shows that model-based optimization can be executed reliably with conventional hardware.

FIGURES

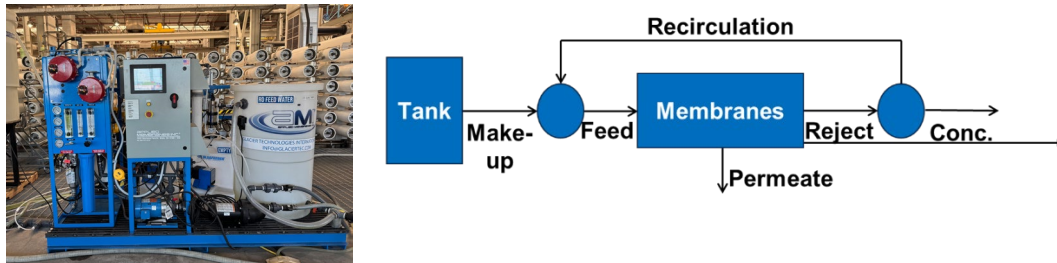


Figure 1. Left: Physical twin of the UF/RO/UV/AOP system. Right: Flow scheme of the RO unit.

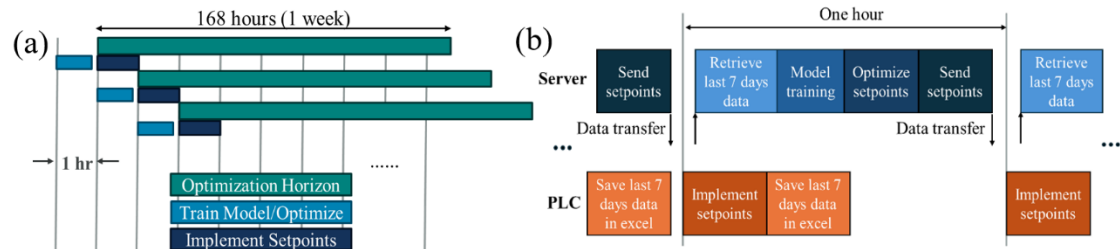


Figure 2. Digital twin control scheme for DT-driven optimization: (a) overall loop, (b) hourly cycle.

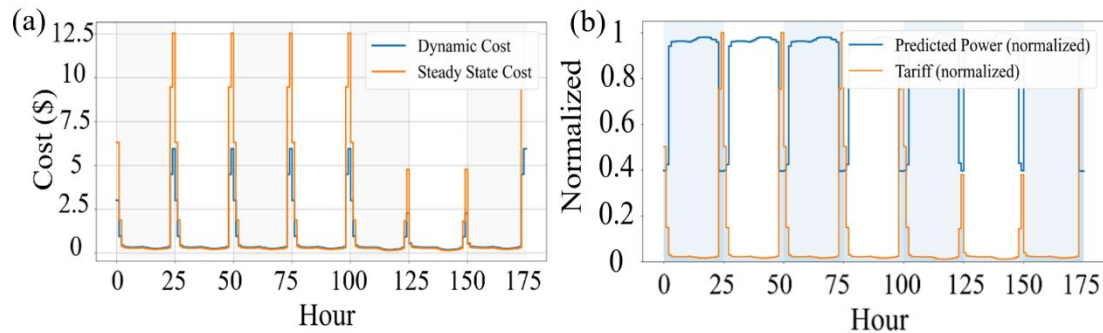


Figure 3: (a) Comparison of electricity cost under nominal steady-state operation and optimized control at the 49th hour of deployment, (b) Predicted power profile overlaid with the tariff structure, demonstrating cost-aware adaptation.

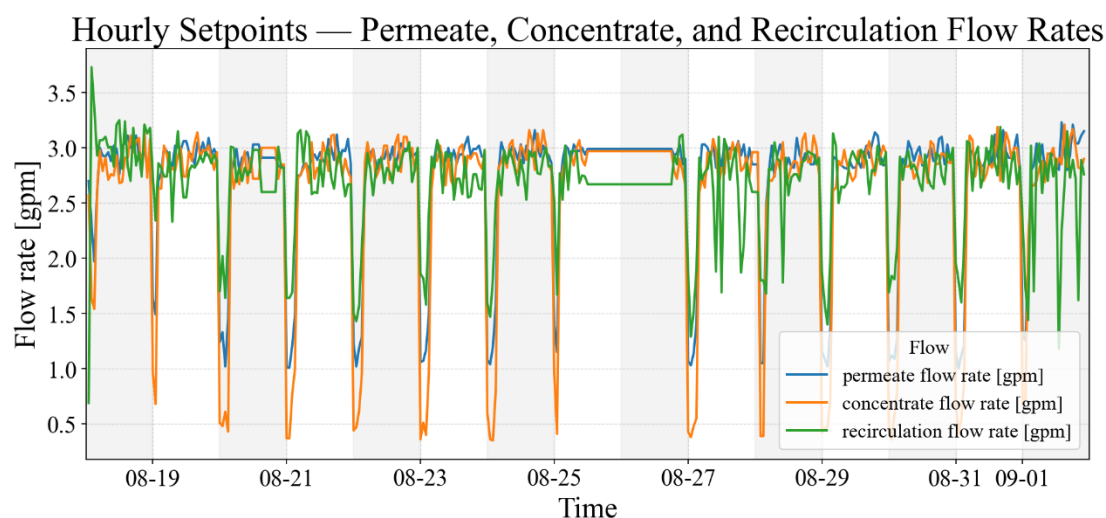


Figure 4: Setpoint applied on an hourly basis on the pilot. The digital twin is capable of reliably updating the flow rate setpoints of the RO unit following hourly updates of the model and setpoint sequence.

REFERENCES

- Conejos Fuertes, P.; Martínez Alzamora, F.; Hervás Carot, M.; Alonso Campos, J. C. Building and exploiting a digital twin for the management of drinking water distribution networks. *Urban Water J.* 2020, 17, 704–713. <https://doi.org/10.1080/1573062X.2020.1771382>
- Fonti, V.; Belitser, E. Feature selection using LASSO. *VU Amsterdam research paper in business analytics* 30 (2017): 1-25.
- Tibshirani, R. Regression shrinkage and selection via the lasso: a retrospective. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 73(3) (2011): 273-282.
- Ghorbani Bam, P.; Rezaei, N.; Roubanis, A.; Austin, D.; Austin, E.; Tarroja, B.; Takacs, I.; Villez, K.; Rosso, D. Digital Twin Applications in the Water Sector: A Review. *Water* 2025, 17, 2957. <https://doi.org/10.3390/w17202957>